

基于 SIG-Multicluster 标准 API 构建多集群 AI

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Part 01

Beyond the Single Cluster



130,000-Node Challenge

Containers & Kubernetes

How Google Does It: Building the largest known Kubernetes cluster, with 130,000 nodes

November 22, 2025

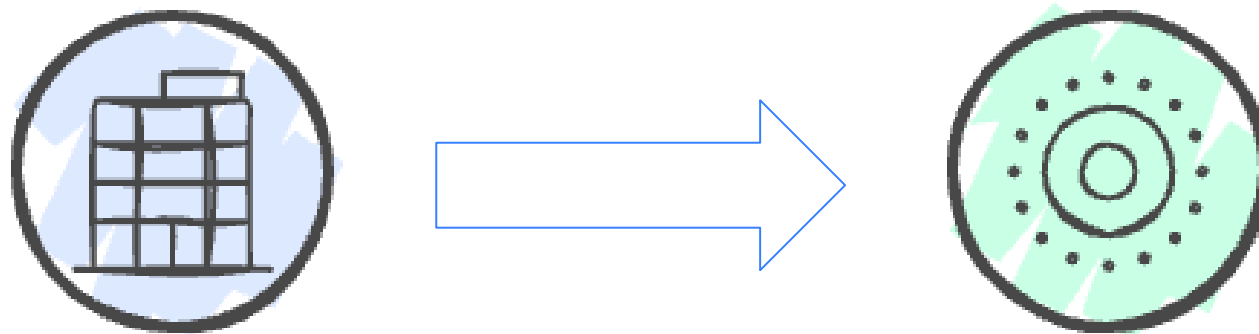
Besher Massri
Software Engineer

Maciek Różacki
Group Product Manager

- The demand for large clusters to stabilize around the **100K node** mark.
- NVIDIA GB200 **$2700W * 10K = 27,000,000W$ (27 兆瓦 (MW))**
- For AI platforms exceeding 100K nodes, need robust multi-cluster solutions that can orchestrate distributed training or reinforcement learning across clusters and data centers.



Beyond the Single Cluster

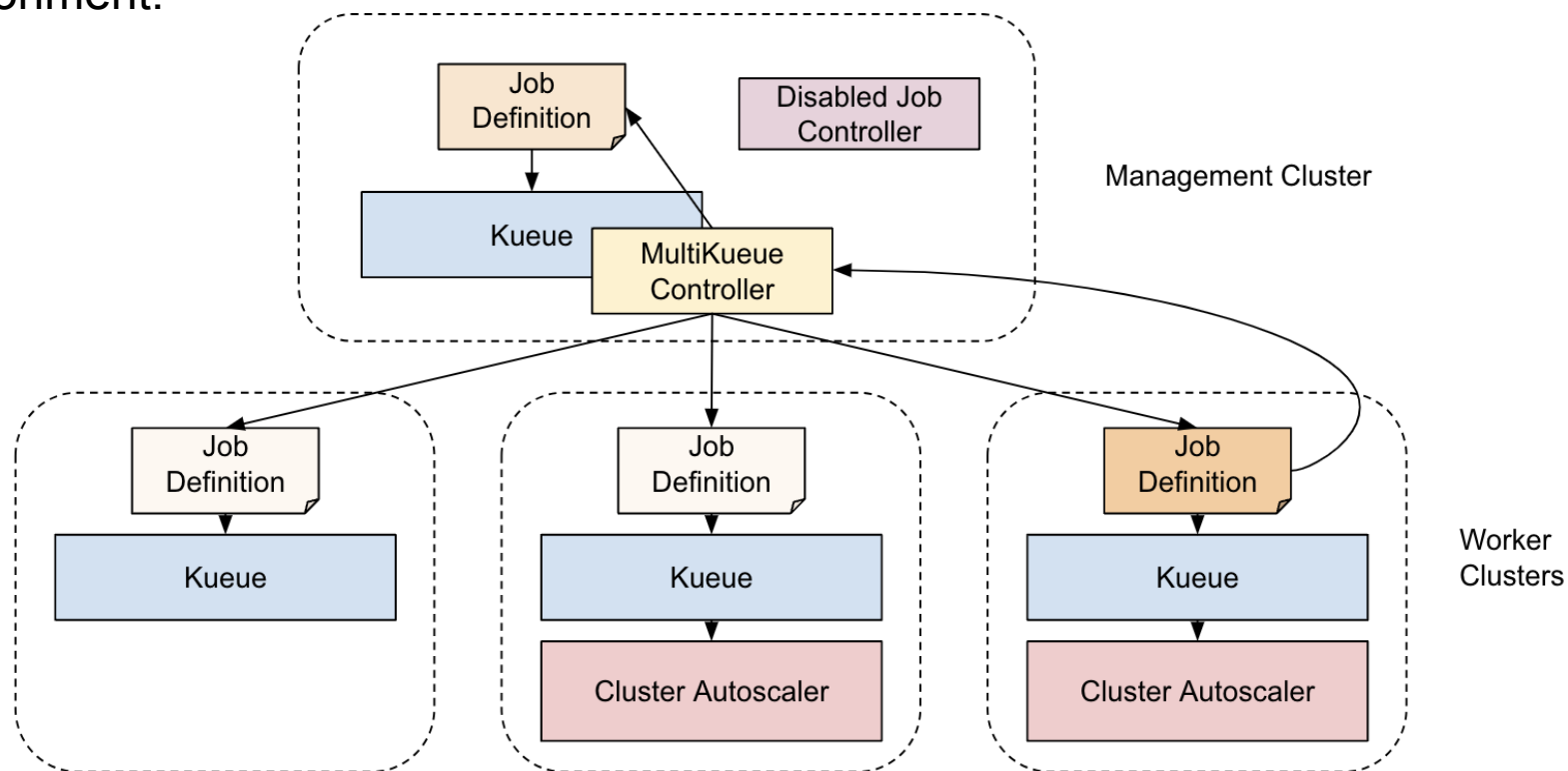


- Power Limit
- Avoid vendor lockin (cloud, on premises and edge)
- Isolation (security and easier management)
- Business unit (each team gets their own cluster)
- Environment separation (development, production)



The Last-Mile Challenges

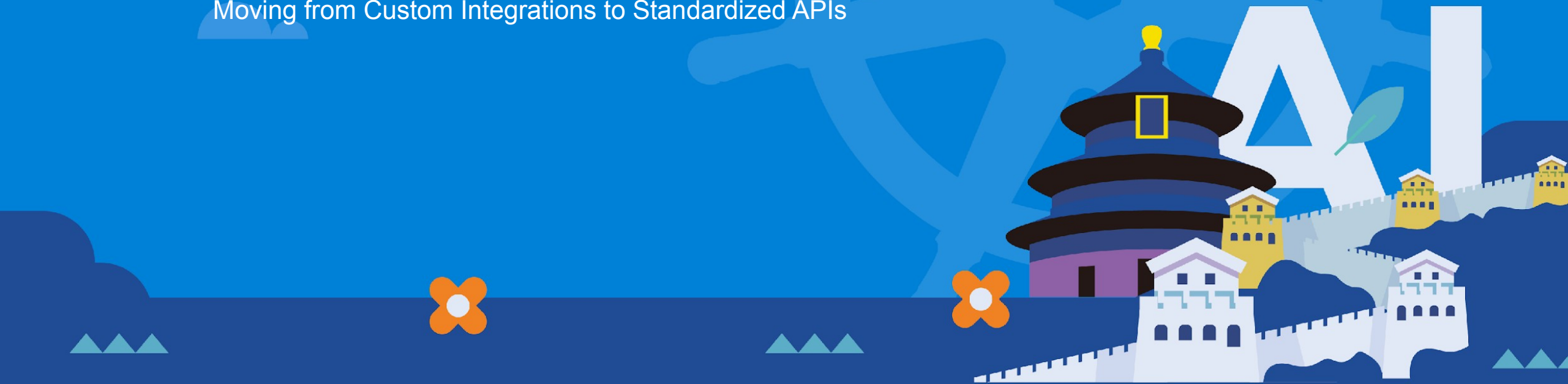
Kubernetes-native Job queue controller open-sourced by the Kubernetes community, with support for multi-cluster environment.



Part 02

SIG-Multicluster Standard API

Moving from Custom Integrations to Standardized APIs



SIG-Multicluster API

ClusterProfile API

An individual member of the Cluster Inventory that details the properties and status of a cluster.

PlacementDecision API

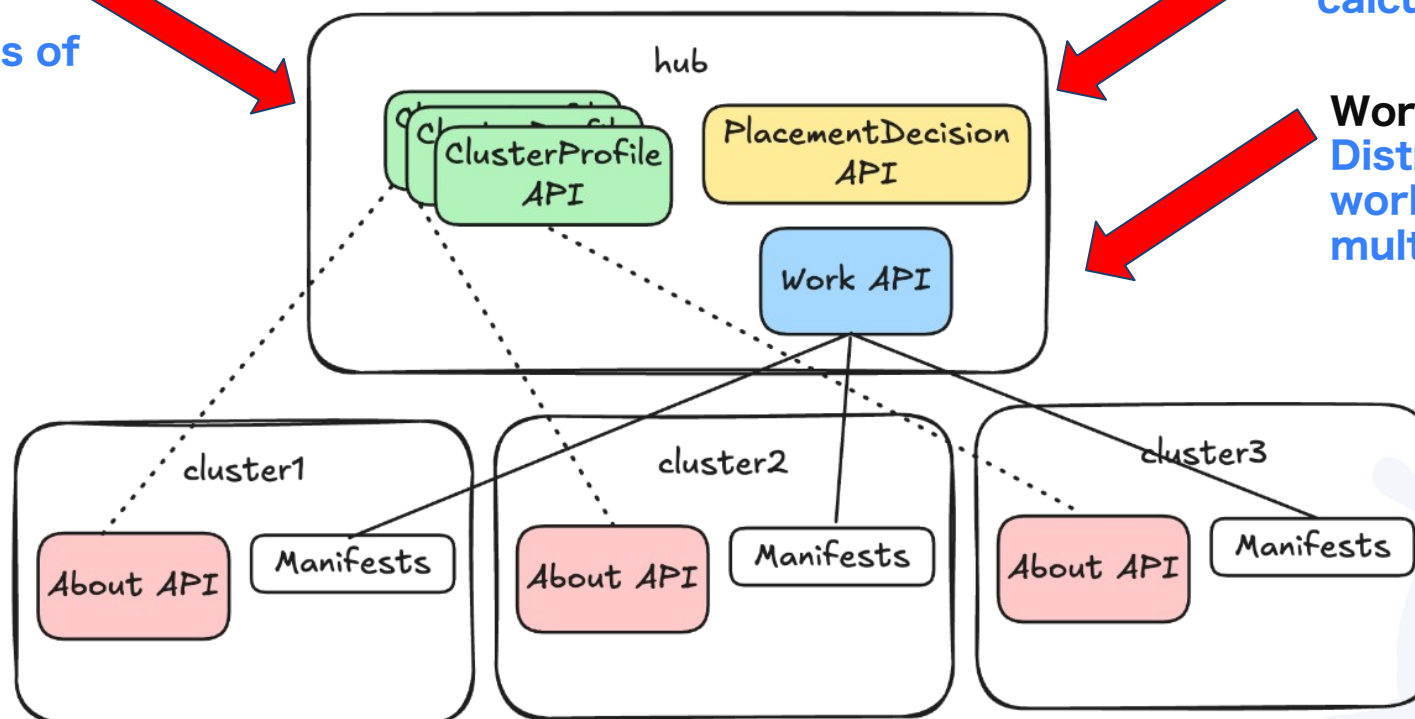
Standardizes the output of multicluster placement calculations.

Work API

Distribute workloads across multiple clusters.

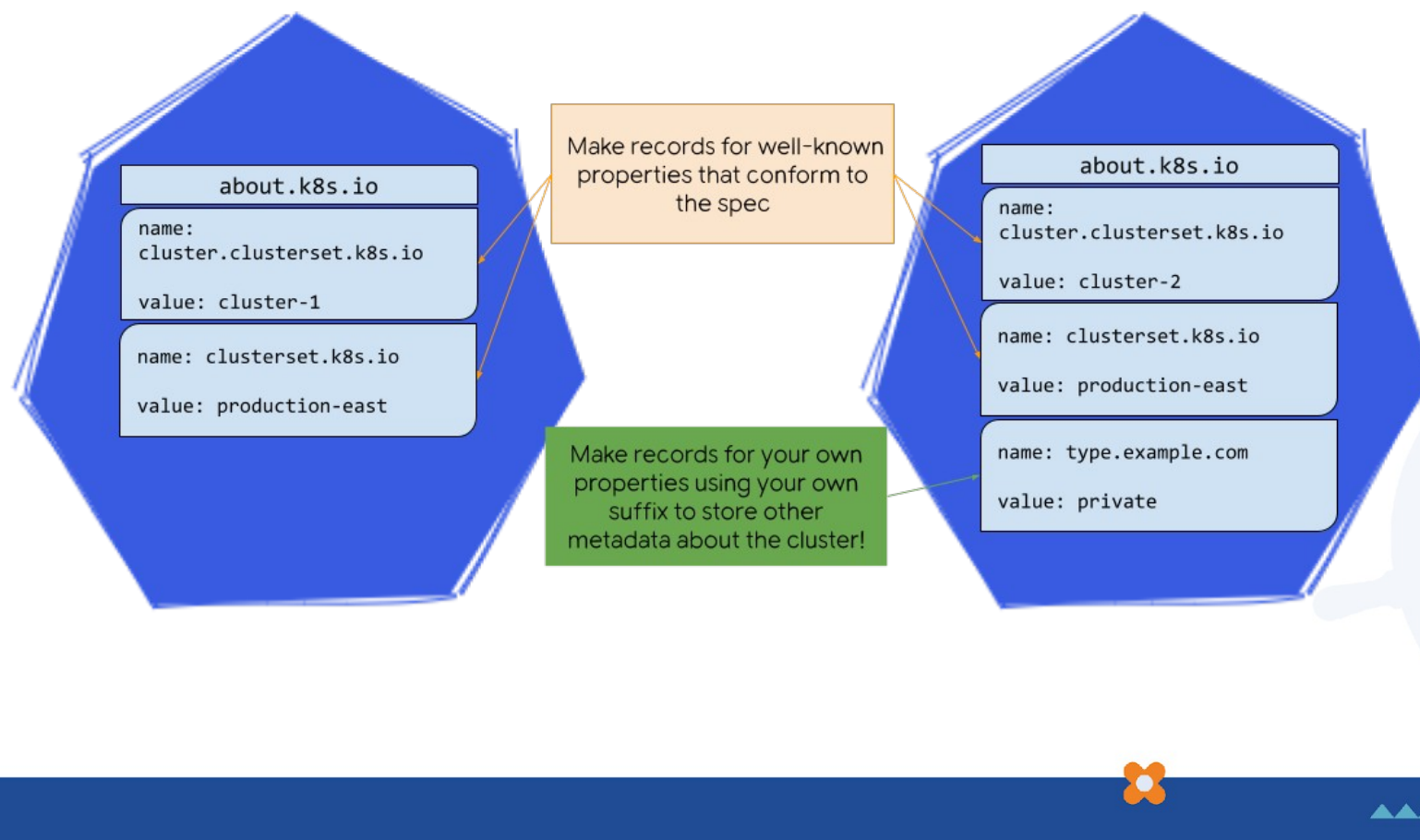
About API

Allows metadata to be attached to individual clusters



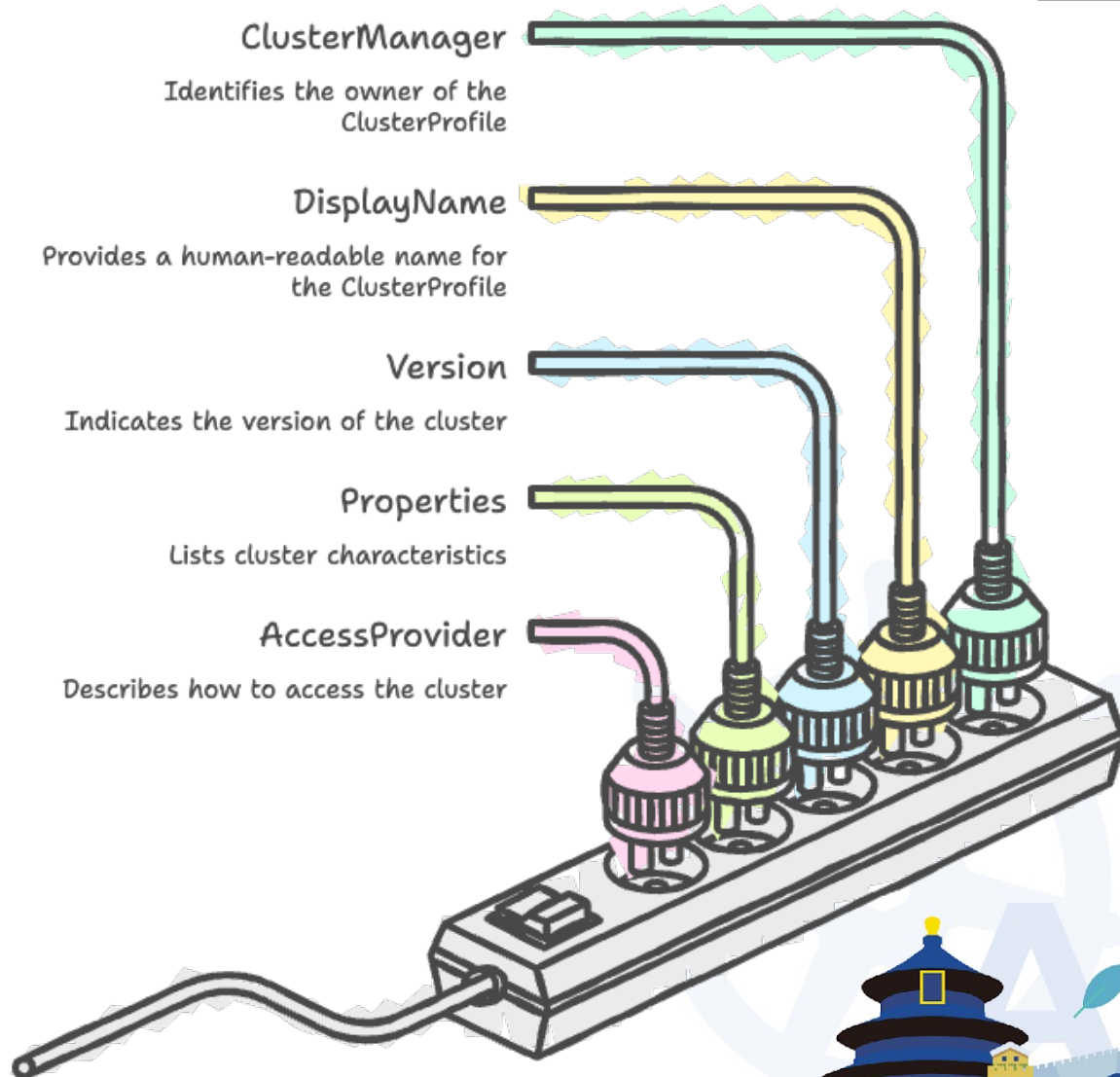
About API

- The About API allows metadata to be attached to individual clusters. The fundamental resource defined in the API is the ClusterProperty CRD.



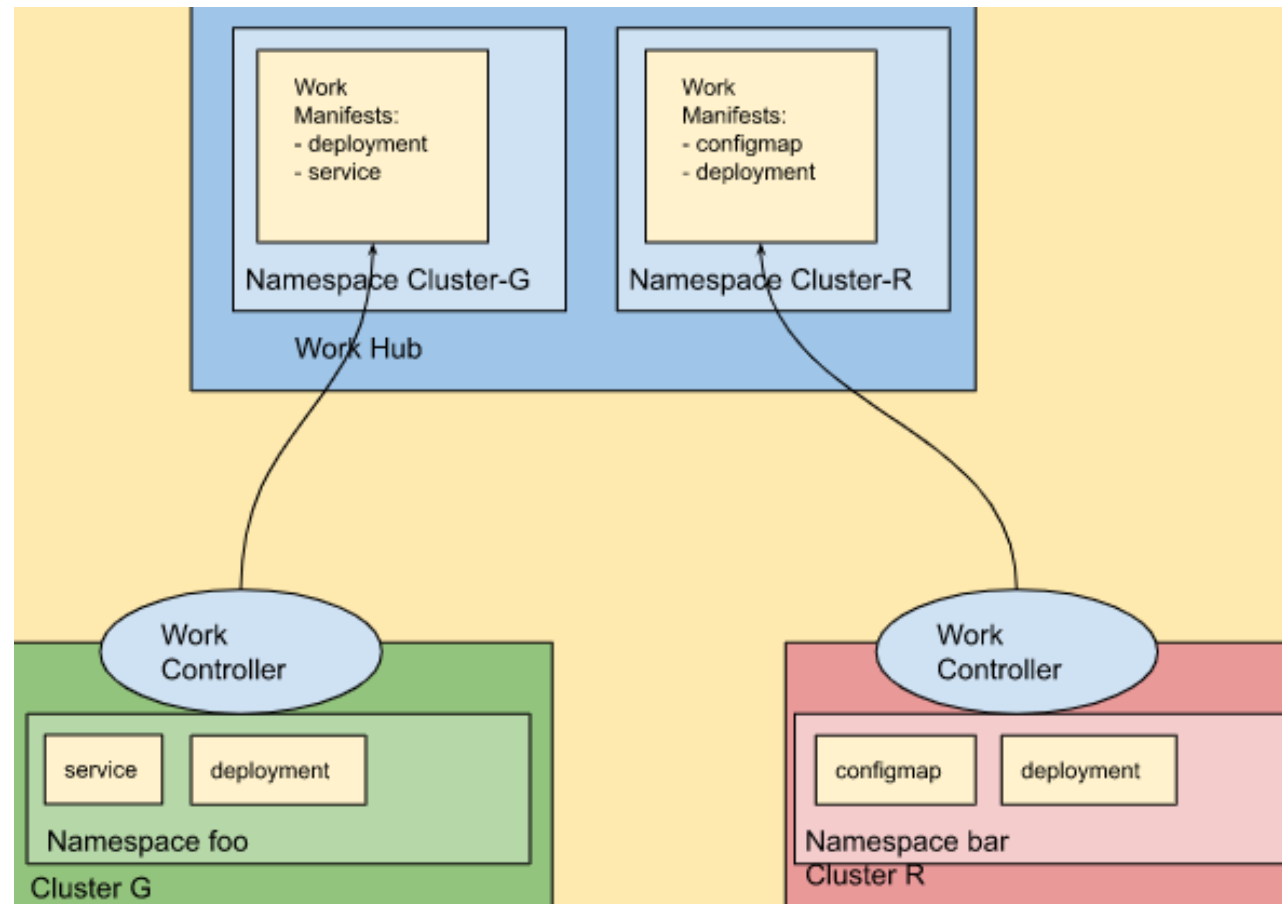
Cluster Profile API

```
apiVersion: multicluster.x-k8s.io/v1alpha1
kind: ClusterProfile
metadata:
  name: some-cluster-name
  namespace: fleet-system
  labels:
    x-k8s.io/cluster-manager: some-cluster-manager
spec:
  clusterManager:
    name: ...
  displayName: some-cluster
status:
  version:
    kubernetes: 1.28.0
  properties:
    - name: clusterset.k8s.io
      value: some-clusterset
    - name: location
      value: apac
  accessProviders:
    - name: secretreader
      cluster:
        server: https://api.some-cluster.example:6443
        certificate-authority-data: <base64-encoded-ca>
        extensions:
          - name: client.authentication.k8s.io/exec
            extension:
              clusterName: some-cluster-name
  conditions:
    - lastTransitionTime: ...
      message: The control plane is of a healthy state
      observedGeneration: 1
      reason: HealthCheckCompleted
      status: True
      type: ControlPlaneHealthy
  ...
```



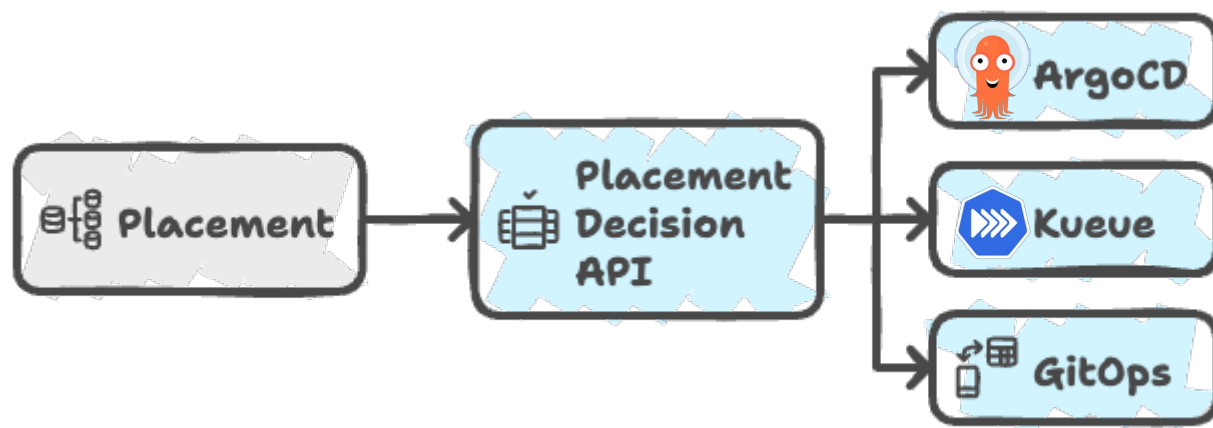
Work API

- A Work is a custom resource that represents a list of API resources to be deployed on a cluster.



Placement Decision API

- Acts as the interface between schedulers and consumers.
Schedulers write decisions using this format;
consumers read from it.



Made with Napkin

apiVersion: multicluster.x-k8s.io/v1alpha1

kind: PlacementDecision

metadata:

name: app-placement-decision-0

namespace: argocd

labels:

multicluster.x-k8s.io/placement-key: "my-app"

multicluster.x-k8s.io/decision-key: "argocd-app-placement-decision"

multicluster.x-k8s.io/decision-index: "0"

schedulerName: multicluster-placement-controller

decisions:

- **clusterProfileRef:**

apiVersion: multicluster.x-k8s.io/v1alpha1

kind: ClusterProfile

namespace: fleet1

name: cluster1

reason: "GPUs available"

- **clusterProfileRef:**

apiVersion: multicluster.x-k8s.io/v1alpha1

kind: ClusterProfile

namespace: fleet1

name: cluster2

reason: "GPUs available"



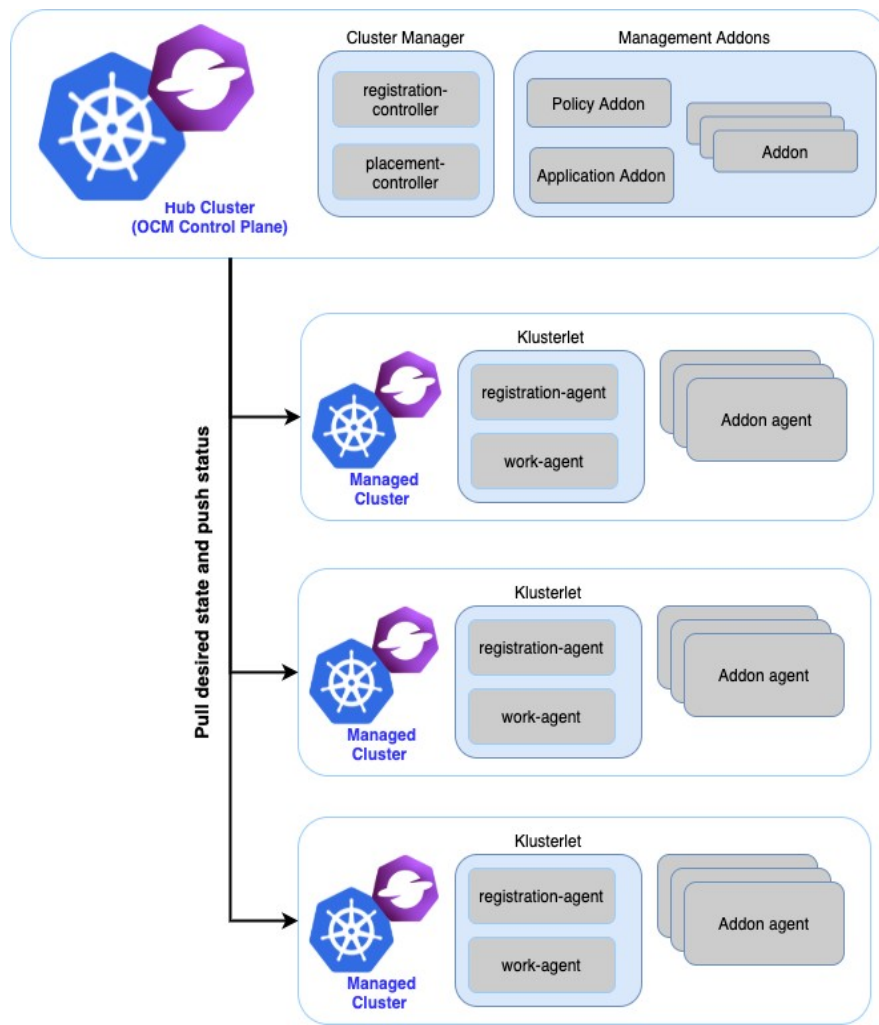
Part 03

Build Multi-Cluster AI with Standard APIs

Access 、 Distribute and Schedule



Open Cluster Management



 **Hub and Spoke Architecture**
A central hub cluster manages multiple spoke clusters.

 **Cluster Inventory**
Register multiple clusters to a hub cluster for management.

 **Work Distribution**
The work API applies resources to managed clusters from the hub.

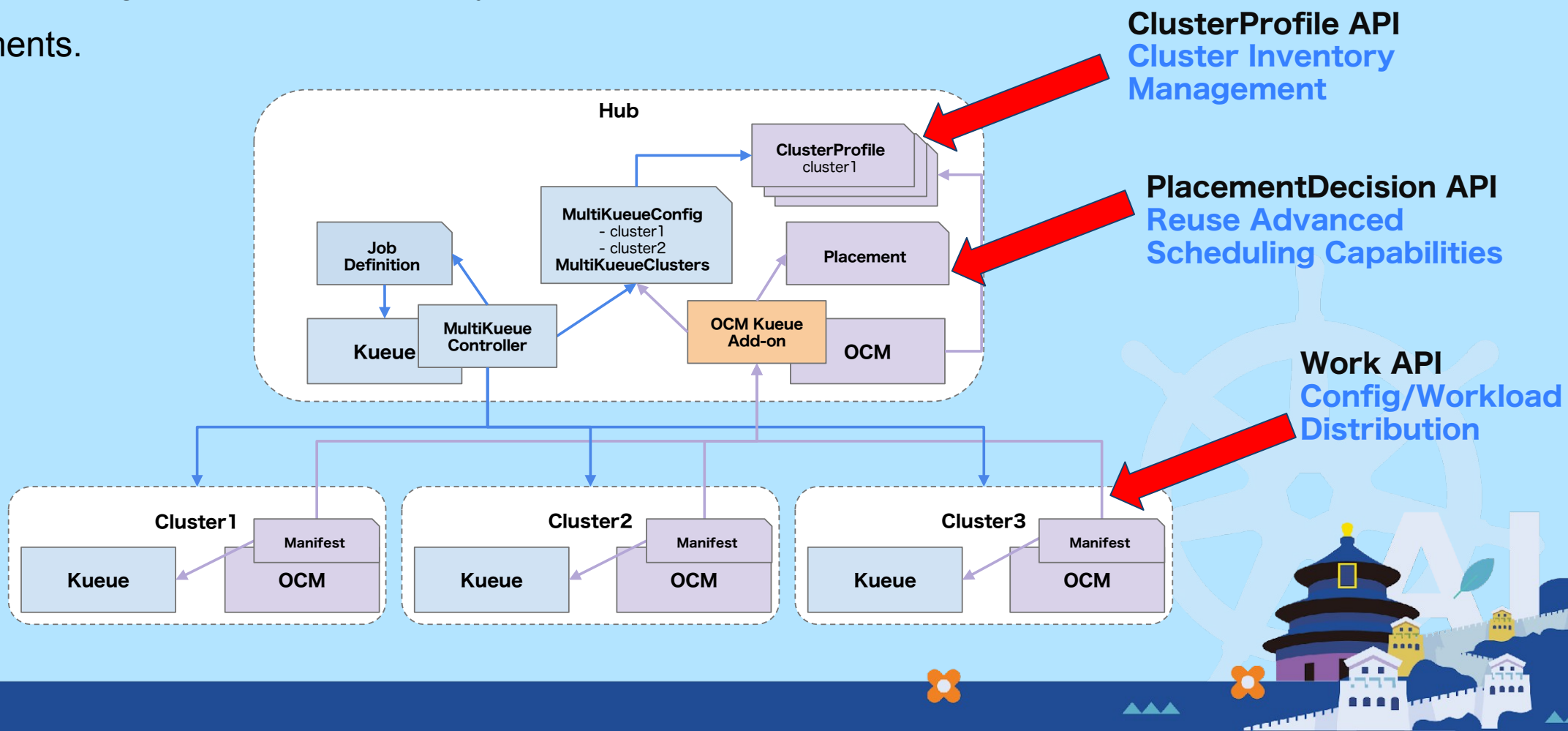
 **Content Placement**
Dynamically place content and behavior across multiple clusters.

 **Add-On Extensibility**
Integrate add-ons to make Kubernetes capabilities multicluster aware.

 **Vendor Neutral API**
Avoid vendor lock-in by using APIs not tied to specific providers.

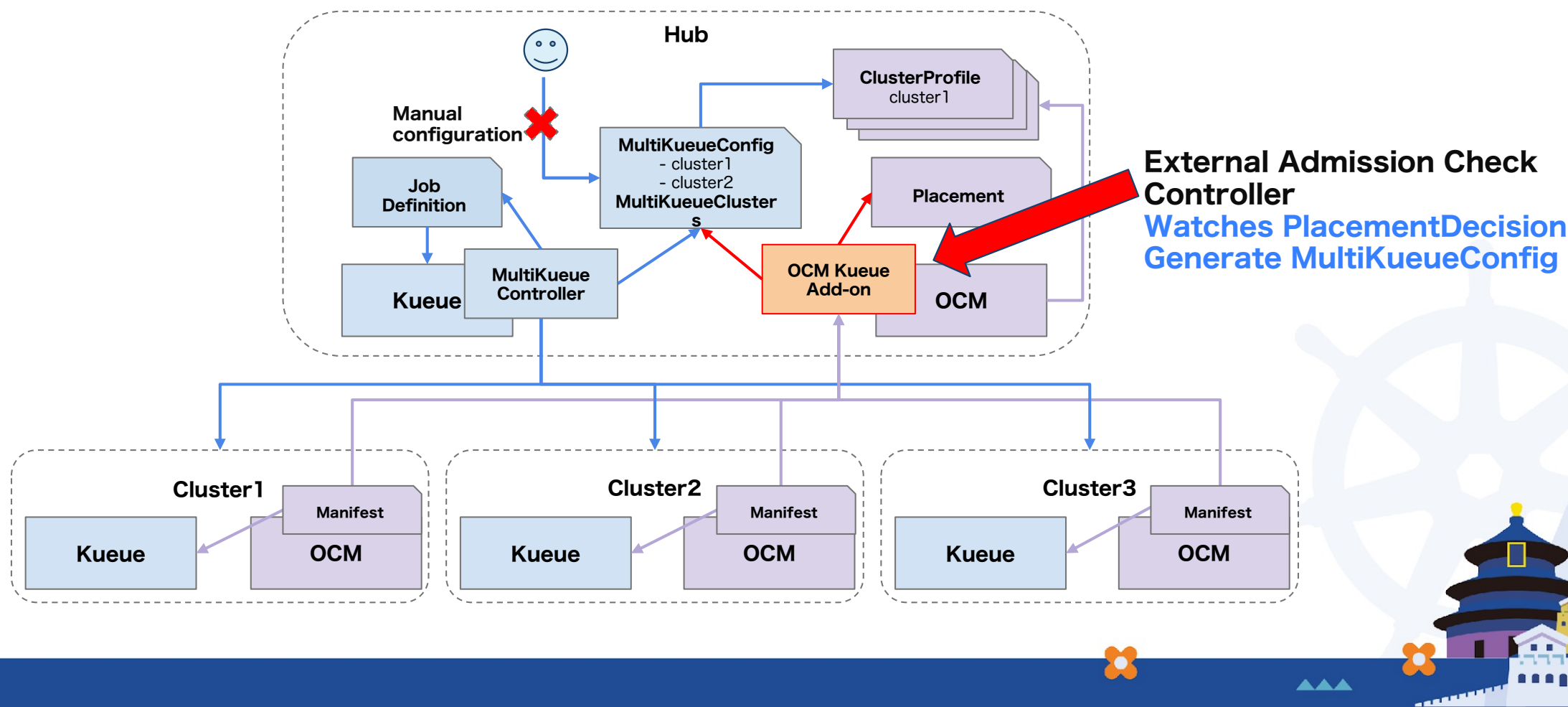
OCM + MultiKueue

- By supporting the SIG-Multicluster standard APIs, such as the ClusterProfile API, it becomes easier to reuse the existing multi-cluster capability in applications like Kueue that support multi-cluster environments.



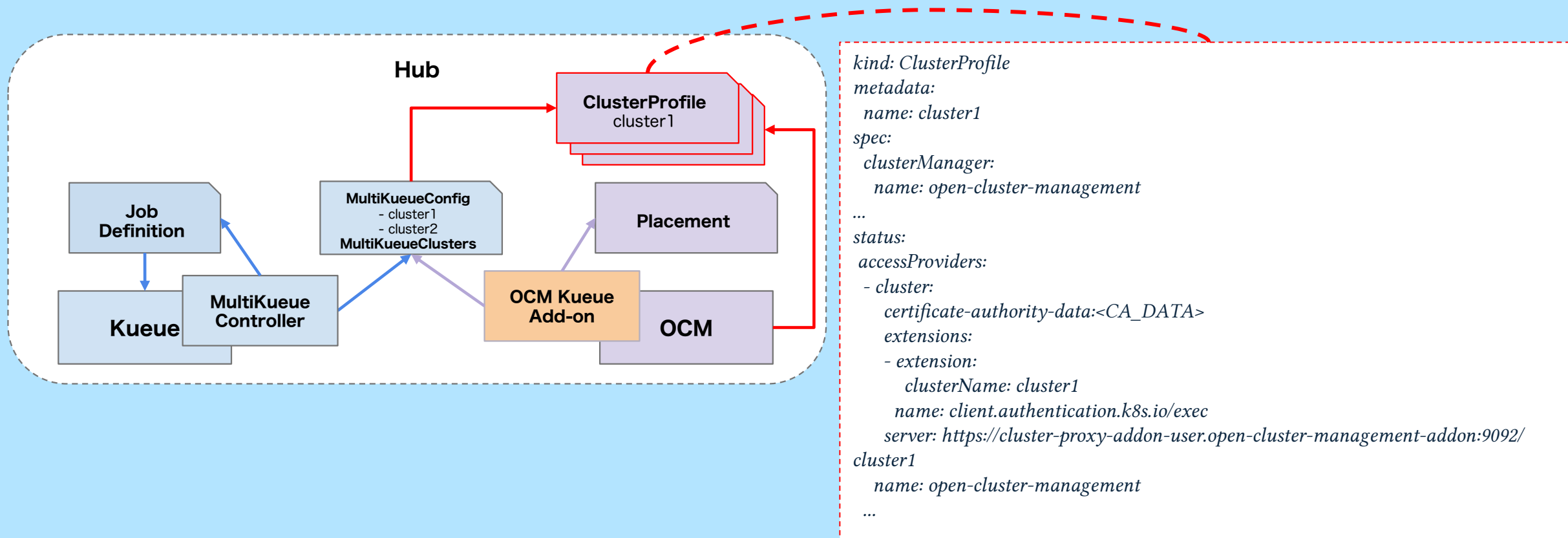
Kueue Add-on and Admission Check

- Admission Check is a mechanism that allows internal or external components to influence the workload's admission.



Access : Cluster Profile API and Access Provider

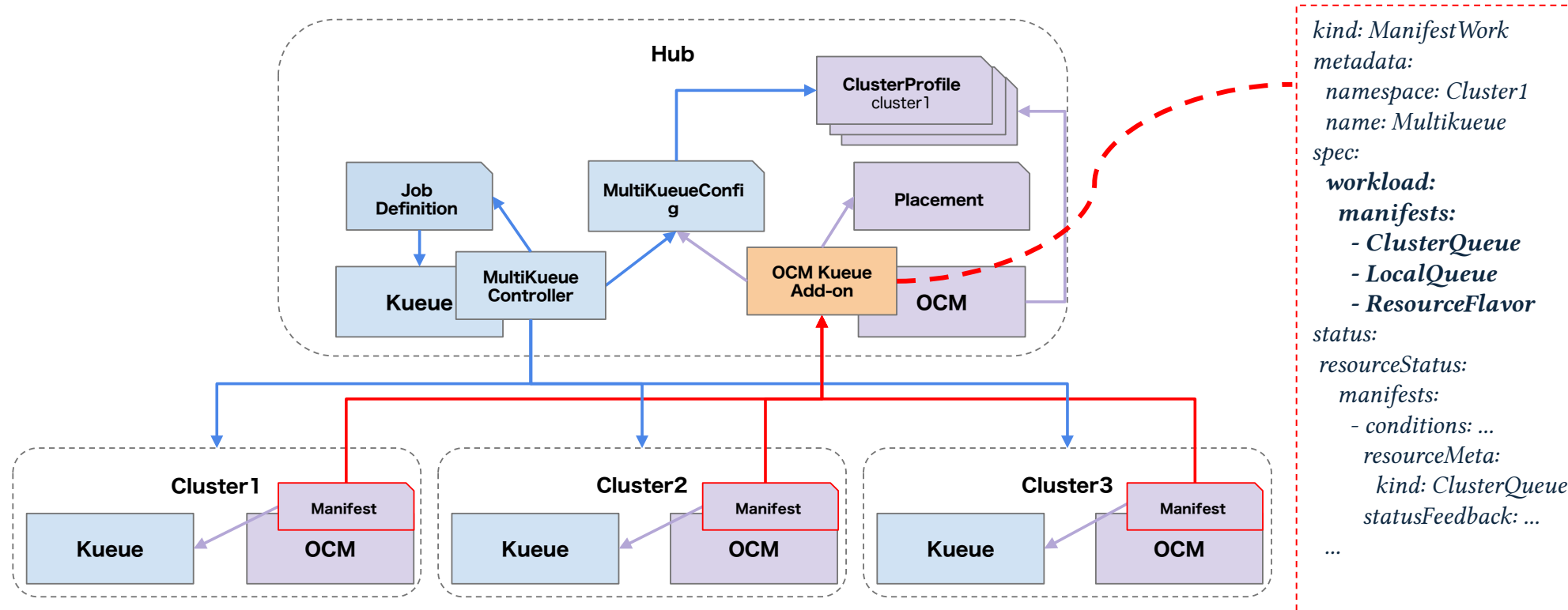
- Kueue now supports reading ClusterProfile when connecting to clusters. The ClusterProfile can be managed by Cloud provider (GKE) and Cluster management platform (OCM).



Support clusterprofile access provider for kueue-addon #103 by @morvencao

Distribute : Work API

- OCM's ManifestWork implements the standard Work API.
- Distribute MultiKueue resources (ClusterQueue, LocalQueue, ResourceFlavor) to hundreds of clusters. Centralized status tracking from the Hub without manual per-cluster access.



Schedule : Placement

- OCM plans to support the standard PlacementDecison API.
- It will be more easier to reuse the existing multi-cluster scheduling capability in applications like MultiKueue, Argo CD. Enabling dynamic configuration and meeting more scheduling scenarios.

Hardware Heterogeneity

Address diverse hardware.

Improving Resource Utilization

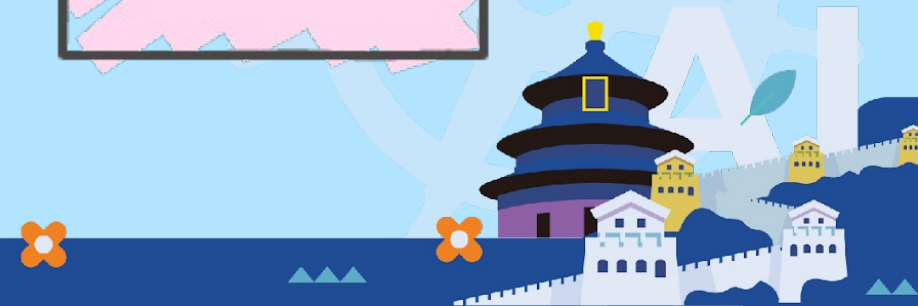
Optimize the use of available resources.

Reduce Resource Fragmentation

Minimize wasted resources due to inefficient allocation.

Dynamic Scoring

Evaluate based on dynamic metrics:
Cost efficiency,
Network latency,
Predictive metrics.



Part 04

Next



Next

- **Dynamic Scoring Framework**

<https://github.com/open-cluster-management-io/addon-contrib/tree/main/dynamic-scoring-framework>

- **Adopting a Standardized PlacementDecision API from SIG Multi-Cluster**

<https://github.com/open-cluster-management-io/ocm/issues/1373>



Get Involved



Open Cluster Management
<https://open-cluster-management.io/>



Thanks.

